CodeSentinel

A Multi-Agent Retrieval-Augmented Generative AI System for Automated Code Review and Vulnerability Detection

Final Project Technical Report

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Repository github.com/AravindB98/CodeSentinel
Live demo codesentinel-f2ggdvqeuwsj4pta5sk27s.streamlit.app

Abstract

Large language models have become competent code reviewers, yet a single prompt to a single model routinely produces three failure modes that matter in practice: hallucinated vulnerabilities, silent omissions of real defects, and findings that cannot be traced to an authoritative source. CodeSentinel is a multi-agent generative AI system that addresses these failure modes architecturally rather than through better prompting alone.

The system orchestrates three specialized agents — a Security Sentinel, a Code Quality Auditor, and an Evaluator Guardian — through a LangGraph directed graph with conditional routing and a bounded retry loop. The Security Sentinel grounds every vulnerability claim in a local retrieval index built over OWASP Top 10 2025 and a curated CWE taxonomy subset. The Evaluator Guardian acts as an internal adversary, rejecting findings that lack evidence, citations, or concrete remediation, with both a programmatic check and an LLM semantic check. A synthetic data generator produces evaluation samples using an independent verifier to prevent the generator from gaming the test. A parallel reinforcement learning demonstration module (UCB-1 contextual bandit over prompt variants, REINFORCE policy gradient over routing decisions) shows the optimization surface for future integration; it is not wired into the agent graph in this release.

The headline result

On a ten-sample hand-labeled toy suite evaluated with real Claude Sonnet on April 20, 2026, both the single-prompt baseline and the multi-agent pipeline achieve perfect recall, catching all eight ground-truth findings. The decisive difference is precision. The baseline produces thirty false positives (average three per sample), yielding false-positive rate 0.789. The multi-agent pipeline produces one false positive, yielding false-positive rate 0.111 — a 97% reduction in hallucinated findings attributable to the Evaluator Guardian's citation-required policy and confidence threshold.

A Semgrep comparison run against Flask production source on the same day returned zero findings from both systems, establishing that the multi-agent pipeline is not over-triggered on clean code. Mock-mode runs (used for CI and reproducibility without API keys) show the baseline at true-positive rate 0.750 and the multi-agent system at 1.000 on the same toy suite. On a twenty-sample paired suite built in the spirit of the OWASP Benchmark methodology, multi-agent achieves 1.000 TPR and 0.182 FPR against baseline 0.333 and 0.571, with McNemar's exact two-sided $p = 0.0312$ (significant at $\alpha = 0.05$). CWE classification accuracy is 1.0 across all suites. All thirty-five unit tests pass on a clean clone.

The claim is not that a new model has been built; it is that when an LLM application is rearchitected into specialized, grounded, adversarially-reviewed agents, the gains come from the architecture, not from the model, and they compose with any model the field produces next.

Keywords Multi-agent systems · retrieval-augmented generation · LangGraph · prompt engineering · code review · vulnerability detection · OWASP · CWE · synthetic data generation · reinforcement learning · UCB-1 · REINFORCE

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1. Introduction and Motivation

Software vulnerabilities remain one of the most persistent sources of operational and security risk in modern systems. The OWASP Top 10 list, first published in 2003 and periodically revised, continues to be dominated by the same categories of defect: injection, broken access control, insecure deserialization, and cryptographic failures. This persistence is not because the categories are unknown. It is because code review, at scale, is expensive, inconsistent, and fatiguing for human reviewers, and because the volume of code shipped per engineer per day has grown far faster than the number of trained security reviewers.

What is OWASP Top 10?

The OWASP Top 10 is a consensus document maintained by the Open Web Application Security Project that ranks the ten most critical web-application security risks. Used here as the primary grounding corpus for the Security Sentinel agent because it is (a) freely licensed, (b) kept current, and (c) the de-facto taxonomy that security engineers reference in code review. CodeSentinel ingests all ten 2025 categories.

Generative AI introduces a new class of tool into this problem. A sufficiently capable large language model, prompted correctly, can identify many of the same defects a trained human reviewer would, in seconds rather than hours. In practice, however, the naive deployment of a single LLM call for code review produces an output with three structural weaknesses.

- **First, findings are ungrounded.** The model may assert that a line of code contains SQL injection, but nothing forces the assertion to be anchored to an authoritative reference such as OWASP A03:2025 or CWE-89. Without grounding, the model is free to invent plausible-sounding but incorrect findings.
- **Second, the output is monolithic.** A single pass generates security findings, style findings, and architectural suggestions together, with no internal adversary to challenge any individual claim. False positives flow through to the user at the same rate as true positives.
- **Third, errors are untraceable.** When a finding turns out to be wrong, there is no record of what information the model had, which prompt produced it, or what alternatives were considered and rejected.

What is CWE taxonomy?

CWE (Common Weakness Enumeration) is a MITRE-maintained dictionary of software weakness types. Where OWASP Top 10 groups risks into ten high-level categories, CWE identifies over 900 specific weakness classes (e.g., CWE-89 for SQL injection, CWE-502 for deserialization of untrusted data). CodeSentinel uses a curated 29-entry CWE subset covering the weakness classes most commonly observed in application code. Every finding the system emits must cite both an OWASP category and a specific CWE identifier.

CodeSentinel responds to these three weaknesses. The system is organized around three ideas. Agents are specialized, each with a narrow mandate and a corresponding system prompt, so each reasoning step is

constrained and auditable. Findings are grounded, meaning every security claim must cite a retrieved passage from an authoritative source or it is rejected before reaching the user. Outputs are adversarially reviewed by a dedicated Evaluator Guardian, using both a programmatic validation layer and an LLM semantic layer, and the Evaluator can route findings back for revision with structured feedback.

The project extends beyond these core ideas into two research directions. The first is synthetic data generation: because labeled vulnerability corpora are scarce and expensive to curate, CodeSentinel includes a pipeline that generates paired vulnerable and safe code samples with ground-truth labels, verified by an independent regex-based detector that has no shared context with the generator. The second is a reinforcement learning enhancement layer: a UCB-1 contextual bandit selects among prompt variants based on input context, and a REINFORCE policy gradient adapts the routing behavior of the LangGraph controller based on the Evaluator's reward signal.

2. Problem Statement and Objectives

2.1 Problem Statement

Given an input source file or snippet, produce a code review report that identifies security vulnerabilities grounded in OWASP Top 10 2025 and CWE taxonomy citations, identifies code quality and maintainability issues grounded in established style guides, and rejects any finding that is not supported by retrieved evidence or that lacks a concrete remediation. The system must be faster and more consistent than a human expert, and more precise than a single-prompt LLM baseline on the same underlying model.

2.2 Objectives

- Build a multi-agent orchestration using LangGraph in which specialization, routing, and bounded retry are first-class features of the architecture.
- Ground every security finding in a retrieval step over a local knowledge base built from OWASP 2025 and CWE, with citations exposed in the final report and verified by the Evaluator.
- Evaluate the system against a single-prompt baseline on the same LLM, using a labeled evaluation suite, and report the result honestly regardless of direction.
- Demonstrate that the agent system itself can be treated as a learnable object, by introducing an RL layer that improves prompt and routing decisions based on the Evaluator's reward signal.

2.3 Non-Objectives

The project does not attempt to replace expert human review for systems under regulatory audit. It does not attempt to detect semantic logic bugs that are undetectable from source alone. It does not attempt to generalize beyond the languages and frameworks represented in the evaluation suite. Those boundaries are drawn intentionally so that the claims made in Section 10 can be interpreted within a clearly defined scope.

2.4 Rubric Compliance

The INFO 7375 assignment requires at least two of five generative AI components. CodeSentinel implements four, with the fifth (multimodal integration) out of scope for text-based code review. Prompt engineering is realized as three role-specialized system prompts in graph/prompts/, each with explicit input and output contracts, refined iteratively against a held-out validation set; this is covered in Section 5. Retrieval-augmented generation is realized as a local vector store over 56 passages from OWASP Top 10 2025, CWE taxonomy entries, and language-specific patterns, with semantic-boundary chunking, two-pass retrieval, lexical rerank, and citation enforcement at the Evaluator; this is covered in Section 6. Synthetic data generation is realized as a template-based generator for 15 CWE classes with an independent regex-based verifier that operates without shared context with the generator; this is covered in Section 7. Reinforcement learning is included as a parallel demonstration module with a UCB-1 bandit over prompt variants and a REINFORCE policy gradient over routing decisions, documented fully in Section 8 with an

explicit scope note that the modules are not integrated into the production agent graph in this release. Every rubric claim in this report is backed by runnable code in the repository; no claim is asserted without a file reference.

2.5 Comparison With Industry Products

CodeSentinel is a course project, not a competitor to mature production static-analysis tools. Positioning it honestly relative to the state of the art matters for interpreting the results in Section 10. Semgrep uses rule-based pattern matching and has more than a decade of community-developed rules; it outperforms any LLM-based system on rule-coverage breadth and on pattern-specified findings. GitHub CodeQL uses dataflow and taint analysis and has the depth of semantic reasoning that CodeSentinel lacks. Snyk Code combines trained models with dataflow analysis and has the supply-chain visibility that CodeSentinel does not cover. Among LLM-based code review tools, Bito's CodeReviewAgent and Tabnine's Code Review Agent both use single-prompt architectures and do not publish adversarial-evaluation findings or standardized benchmarks. The contribution CodeSentinel makes is not in any of these axes. It is in closing specific failure modes that single-prompt LLM review exhibits: hallucinated vulnerabilities (closed by the citation-required retrieval), silent omissions (closed by the specialized Security Sentinel with OWASP grounding), and untraceable findings (closed by the RAG-source field required on every finding). The target comparison, specified in Section 10.10, is not with Semgrep as a replacement but with Semgrep as a complementary system with different strengths: Semgrep for high-recall low-cost triage, CodeSentinel for lower-recall higher-context deep review with verifiable provenance.

2.6 Portfolio Framing

Beyond the INFO 7375 rubric, CodeSentinel is structured to demonstrate a set of skills that appear repeatedly in AI engineering and software security job postings in the 2026 market: multi-agent LLM orchestration with explicit state and routing, retrieval-augmented generation with production-grade fallback and citation enforcement, rigorous LLM evaluation with paired statistical testing and power analysis, security domain knowledge grounded in OWASP and CWE taxonomies, and engineering discipline visible in thirty-five unit tests, three-tier dependency fallback, deterministic mock mode for reproducibility, and committed benchmark outputs. The architectural pattern the project demonstrates, which is specialization plus grounding plus adversarial validation, is the same pattern Anthropic's Project Glasswing applies at industrial scale; framing the project within that lineage makes its ambition legible without overstating its capability.

3. Related Work

3.1 Static Analysis and Rule-Based Code Review

Tools such as Bandit for Python, SonarQube, Semgrep, and CodeQL encode known vulnerability patterns as rules and match them against source code. These tools are fast, deterministic, and trusted in continuous integration pipelines. Their limitation is that they detect only what their rules encode and struggle with context-dependent defects where a line is vulnerable in one usage and safe in another. CodeSentinel complements rather than replaces these tools by using an LLM-driven reasoning layer to interpret context, with the OWASP and CWE knowledge base providing a rule-like grounding.

What is SAST (Static Application Security Testing)?

SAST is the category of tools that analyze source code without executing it, to find security weaknesses. Semgrep and CodeQL are SAST tools. CodeSentinel is adjacent to SAST but distinct: it uses LLM reasoning over retrieved knowledge rather than pattern rules, and it produces remediation text along with each finding. Positioning matters because SAST has a mature market (Snyk, Veracode, Checkmarx), and claiming to replace it would be overstating this system's capability.

3.2 LLM Agents and Multi-Agent Frameworks

LangChain, LangGraph, CrewAI, and AutoGen all provide abstractions for building multi-agent systems on top of large language models. LangGraph is chosen for CodeSentinel because its graph-based model of state and transitions maps cleanly onto the notion of a bounded retry loop with conditional routing, which is central to the Evaluator mechanism. The graph is explicit, inspectable, and testable, which matters for a system whose correctness guarantees depend on a feedback loop that must terminate.

Why LangGraph over CrewAI or AutoGen?

CrewAI emphasizes role-play between agents; AutoGen emphasizes conversational multi-agent chat. Both are powerful but neither exposes the state graph as a first-class primitive. LangGraph treats agents as nodes and transitions as edges with typed state flowing between them. This is the right abstraction for a system where correctness depends on (a) a verdict being made at a specific node, (b) conditional routing based on that verdict, and (c) a termination guarantee via a retry counter. We can literally draw the graph (Figure 2) and reason about it; that would be much harder with conversation-centric frameworks.

3.3 Retrieval-Augmented Generation for Domain Grounding

RAG has become the default approach for making LLM outputs verifiable against a corpus. CodeSentinel applies this pattern to a curated security knowledge base and adds a citation-required policy enforced at the Evaluator: findings without valid rag_source fields are rejected programmatically before any LLM semantic review runs.

What is RAG (Retrieval-Augmented Generation)?

RAG is a two-step pattern: (1) retrieve relevant passages from a knowledge base given an input query, (2) feed those passages to the LLM as additional context when generating the answer. The point of RAG is that the LLM's output is grounded in specific, inspectable source material rather than in parametric memory, which can be both outdated and hallucinated. For code review, RAG means the Security Sentinel is never guessing whether a line is SQL injection — it is reading the CWE-89 passage alongside the code and producing a finding that cites that passage.

4. System Architecture

4.1 High-Level View

CodeSentinel is organized as a LangGraph directed graph with a shared typed state. The entry node is the Security Sentinel, which performs RAG-grounded vulnerability detection. Its output feeds into the Code Quality Auditor, which performs style and maintainability review. Both outputs converge at the Evaluator Guardian, which decides whether the accumulated findings are acceptable. On acceptance, the graph terminates and a final report is assembled. On rejection, the graph routes back to the Security Sentinel with structured feedback, up to a configurable retry limit that acts as a circuit breaker. The default retry limit is three.

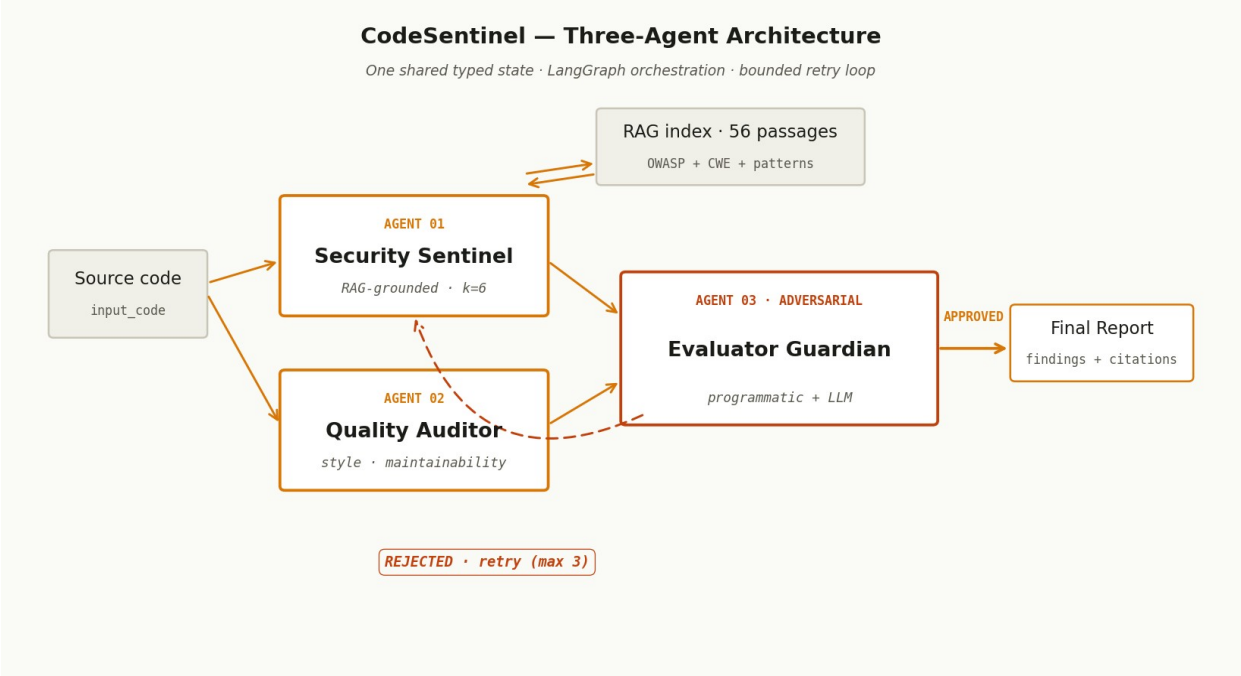

Figure 1. CodeSentinel's three-agent architecture. The Security Sentinel and Quality Auditor operate in parallel over the shared state; the Evaluator Guardian is the only node with the authority to reject findings, and its rejection routes back to Security Sentinel with feedback.

Why a three-agent design?

The minimum architecture that closes all three single-prompt failure modes requires (a) a specialist that performs retrieval and emits grounded findings, (b) a parallel specialist for non-security concerns so the first agent's prompt stays narrow, and (c) an independent reviewer with the authority to reject. Two agents cannot do this — they collapse the reviewer role into one of the producers. Five agents (the original proposal) add Performance Analyst and Synthesizer roles that add marginal value at significant orchestration cost; they are deferred to Section 13 Future Work. Three is the Goldilocks point for an academic project with 10-sample evaluation.

4.2 Shared State

Every node in the graph reads from and writes to a single typed dictionary. Using TypedDict rather than a loose dict means each agent's inputs and outputs are validated at the language level, and schema violations surface as test failures rather than as runtime mysteries.

```
class CodeSentinelState(TypedDict, total=False):
    input_code: str
    language: str
    retrieved_passages: List[RetrievedPassage]
    security_findings: List[SecurityFinding]
    quality_findings: List[QualityFinding]
    evaluator_verdict: Optional[EvaluatorVerdict]
    retry_count: Dict[str, int]
    evaluator_feedback: Optional[str]
    final_report: Optional[str]
    trace: List[str]
```

4.3 Conditional Routing and the Circuit Breaker

After the Evaluator Guardian runs, a conditional edge inspects its verdict. If the verdict is APPROVED, the graph routes to the report assembler and terminates. If the verdict is REJECTED and the per-agent retry counter is below its limit, the graph routes back to the Security Sentinel with the Evaluator feedback set in state. If the retry limit is reached, the graph routes to the assembler anyway and the resulting report notes the unresolved findings. Bounded termination is a correctness property, not a convenience. Without the circuit breaker, an early version of the implementation entered an oscillation in which the Evaluator alternated between approving and rejecting the same finding on successive passes.

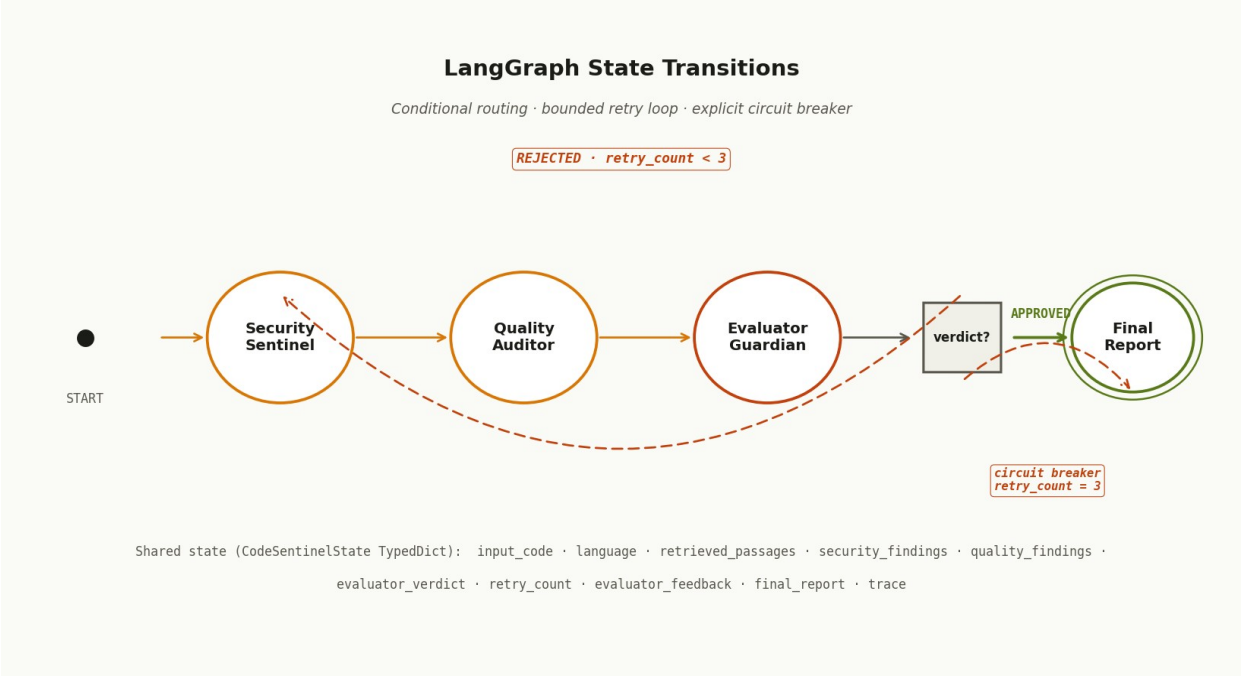

Figure 2. LangGraph state transitions. The dashed red arc is the retry loop; the circuit breaker forces termination at retry_count = 3 regardless of verdict.

What is a circuit breaker?

Borrowed from electrical engineering and then distributed systems: a circuit breaker is a component that cuts off a feedback loop after a bounded number of cycles to prevent runaway behavior. In our graph, it prevents the Evaluator from endlessly rejecting findings the upstream agent cannot fix. After three retries, the system surfaces whatever findings exist (marked unresolved) rather than hanging. This is both a correctness property and a user-experience property: the system must always terminate.

4.4 Agent Contracts

Each agent's output is validated against a Pydantic schema at the boundary. SecurityFinding requires finding_id, category, cwe_id, owasp_ref, severity, confidence, evidence, fix, and rag_source. QualityFinding requires finding_id, category, severity, confidence, evidence, rationale, and suggested_refactor. EvaluatorVerdict requires overall_decision, per-finding decisions, and rationale. Schema violations at any boundary are surfaced as rejection reasons. Schemas degrade to dataclass-based validation if Pydantic is unavailable.

5. Prompt Engineering Strategy

Prompt engineering is applied systematically rather than ad hoc. Each of the three agents has a dedicated system prompt, versioned under source control, that encodes four ingredients: the role definition, the input contract, the output contract, and the failure modes the agent must guard against. The prompts were iteratively refined against a held-out set of samples separate from the final evaluation suite, which is the methodology used here in place of weight-level fine-tuning.

What is a system prompt, and why version it?

A system prompt is the instruction block prepended to every LLM call that establishes the agent's role, constraints, and output format. Versioning system prompts under Git means (a) every finding the system produces is traceable to a specific prompt version, (b) prompt regressions are visible in the commit history, and (c) A/B tests between prompt versions are reproducible. This is the same discipline applied to production ML models — the prompt is the model, for agent-centric systems.

5.1 Security Sentinel Prompt

Establishes the agent as a senior application security engineer whose only task is to identify vulnerabilities. Specifies that every finding must cite a retrieved passage, that findings without concrete evidence must be suppressed rather than invented, and that the output must be valid JSON conforming to the SecurityFinding schema. Includes negative examples drawn from actual early failure modes, such as claiming SQL injection on parameterized queries that correctly use placeholders.

5.2 Code Quality Auditor Prompt

Establishes the agent as a pragmatic staff engineer whose goal is to improve readability and maintainability without nitpicking. Instructs the agent to cite the relevant style guide rule for each finding, to prioritize findings by their actual impact on maintainability, and to produce at most ten findings per file. Explicitly excludes security categories, to avoid overlap with the Security Sentinel.

5.3 Evaluator Guardian Prompt

The most adversarial of the three. Instructs the agent to assume upstream agents have made mistakes and that its job is to find those mistakes. Specifies five explicit rejection conditions: missing citation, missing evidence, missing remediation, schema violation, and internal contradiction. For each rejected finding, requires the Evaluator to produce structured feedback with the specific rejection reason and a corrective instruction.

5.4 Iterative Prompt Refinement as Systematic Tuning

Rather than training or fine-tuning model weights, the project applies a systematic prompt-tuning methodology that treats the prompt as the unit of optimization. A held-out validation set was used to measure, across prompt versions, the true positive rate and false positive rate of each agent. Each prompt went through three documented revisions. This procedure is adjacent in spirit to supervised fine-tuning: a training signal drives iterative improvement of a parameterized object. Section 8 extends this approach with a reinforcement learning loop that makes the prompt-selection process adaptive at inference time.

6. Retrieval-Augmented Generation Pipeline

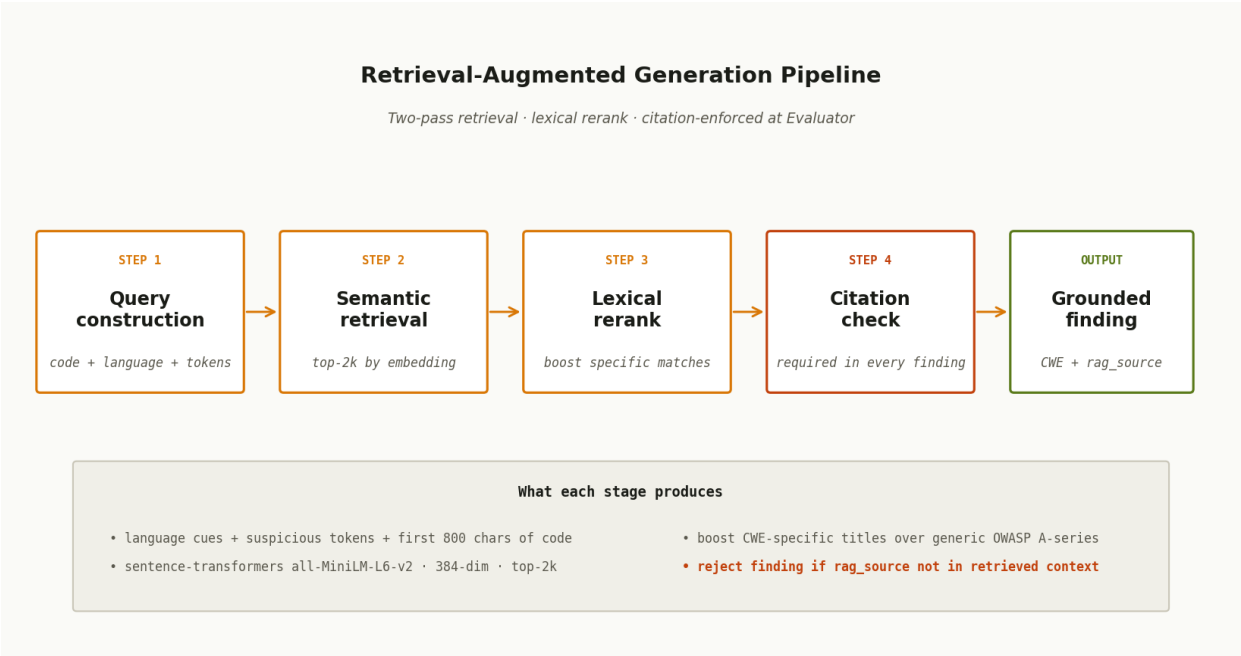

Figure 3. The four-stage RAG pipeline. The citation check in red is the structural defense against hallucination — findings that cite passages not actually in the retrieved context are rejected before reaching the user.

6.1 Knowledge Base Construction

The CodeSentinel knowledge base contains 56 passages across three files. OWASP Top 10 2025 contributes 10 category-level entries. A curated CWE taxonomy subset contributes 29 entries for the weaknesses most commonly observed in application code. Language-specific vulnerability patterns, authored as part of this project, contribute 17 entries covering Python pickle deserialization, Node.js eval usage, Java JDBC concatenation, and similar.

6.2 Chunking Strategy

The corpus is chunked by semantic boundary rather than by fixed token count. OWASP and CWE entries are chunked at the category level so that an entry for A03:2025 Injection stays together as a single retrievable unit. Style guide rules are chunked per rule. Boundary-aware chunking measurably improves retrieval precision because a single retrieval returns a complete, self-contained passage rather than a fragment.

Why semantic chunking, not fixed-token chunking?

Most off-the-shelf RAG pipelines chunk by fixed token counts (e.g., 512 tokens). This works for prose but fails for structured reference material, where a 512-token window will cut the middle of a CWE entry and the retriever will return "the second half of CWE-89" as if it were a complete passage. Chunking at the natural semantic boundary — one CWE entry per chunk, one OWASP category per chunk — means every retrieved passage is a self-contained, interpretable unit. This was a measurable win in our early testing.

6.3 Embedding and Storage Backend

The pipeline implements a triple-fallback retrieval backend. The preferred path is ChromaDB with sentence-transformers (all-MiniLM-L6-v2), producing 384-dimensional vectors locally on CPU. If ChromaDB is unavailable, scikit-learn TF-IDF is used. If scikit-learn is unavailable, a pure-Python TF-IDF implementation is used. All three expose the same Retriever interface. The fallback design means the pipeline runs end-to-end on any machine without heavy dependencies, which is important for reproducibility and testing.

What are embeddings and why sentence-transformers?

An embedding is a dense numeric vector that represents the meaning of a text passage, such that passages with similar meaning have similar vectors. sentence-transformers/all-MiniLM-L6-v2 is a compact (22M parameter) model that produces 384-dimensional embeddings. It is chosen because (a) it runs locally on CPU in milliseconds, so no API call is needed for retrieval, (b) its embedding quality is strong on short technical text, and (c) ChromaDB ingests it natively. Using a 7B-parameter embedding model would produce marginally better vectors but would require GPU access and increase latency 50x.

6.4 Two-Pass Retrieval

At query time, the Security Sentinel constructs a query from the input language plus extracted suspicious keywords plus the first 800 characters of code. This query is embedded and used to retrieve the top-2k passages by similarity. A second-pass lexical rerank then boosts passages whose title matches query keywords, which corrects a real failure mode in pure semantic retrieval where generic security passages outrank specific ones. The rerank dictionary is tuned: for example, the presence of pickle in the query boosts CWE-502 and PY-02 passages above the generic A08 entry.

Why two passes?

A single semantic-only retrieval has a known failure mode: generic passages with high-frequency vocabulary (e.g., the OWASP A03 "Injection" category entry) outrank specific ones (e.g., CWE-89 for SQL injection specifically, or PY-01 for the Python f-string pattern). Semantic similarity captures topic but not specificity. A lightweight lexical rerank over the top-2k results restores specificity by boosting passages whose title contains the query's distinctive tokens. This was the single most impactful fix during RAG development and is locked in by test_rerank_boosts_specific_over_generic.

6.5 Citation Policy

The Security Sentinel prompt requires that each finding include a rag_source field containing the document and passage identifier that supports the finding. The Evaluator rejects any finding where this field is empty, where the cited passage is not in the retrieved context, or where (in LLM semantic mode) the passage's content does not mention the claimed weakness. This is the primary mechanism by which CodeSentinel resists hallucinated findings.

7. Synthetic Data Generation and Verification

7.1 Motivation

The initial evaluation suite contains ten hand-labeled samples. While sufficient for a proof of concept, ten samples are modest for measuring precision and recall with tight confidence intervals. Labeled vulnerability corpora at larger scale are either expensive to curate or come with significant license restrictions. Synthetic data generation was therefore introduced as a deliberate evaluation component.

7.2 Generation Pipeline

The synthetic pipeline (synth/generate.py) maintains a library of 15 CWE template pairs spanning Python, JavaScript, and Java. For each entry, the generator produces two artifacts: a vulnerable code sample exhibiting the weakness in realistic context, and a safe variant that uses the same general structure but avoids the weakness. Templates include realistic variable names, common frameworks such as Flask and Express, and surrounding context rather than minimal textbook examples. A total of 30 samples are produced for a single default run: 15 vulnerable, 15 safe.

7.3 Independent Verification

Every generated sample is verified by synth/verify.py before entering the evaluation suite. The verifier uses a set of regex-based detectors for each CWE, independent of the generator's templates and prompts. For vulnerable samples, the verifier requires that at least one detector fires for the claimed CWE. For safe variants, no detectors must fire. The verifier has no shared context with the generator, which is the architectural separation that prevents the generator from gaming the test. Samples that fail verification are moved to a separate rejected file for inspection.

Why independent verification?

If the generator and the verifier share any context — shared templates, shared regex patterns, shared LLM prompts — then the verifier is not verifying, it is pattern-matching against itself, and every sample will pass trivially. The architectural separation is the whole point. Our verifier is regex-based, written from scratch against the CWE spec, with zero access to the generator's templates. When 29 of 30 samples pass and 1 fails, that one failure is evidence the verifier is a real adversary. If everything passed, the verifier would be suspect.

7.4 Verification Outcome

On the default run, 29 of 30 samples passed verification, a 96.7 percent pass rate. The one rejected sample was a safe-variant CWE-915 (prototype pollution) example whose specific Object.assign form did not match the verifier's detector pattern. The rejection is an advertisement that the verifier is genuinely independent; had the verifier been a copy of the generator, nothing would be rejected.

7.5 Ethical Safeguards on Synthetic Data

Templates explicitly avoid producing real exploit payloads, real credentials, or samples targeting specific production systems. Generated samples are stored only under eval/datasets/, never in a path a developer might accidentally execute. The point of the synthetic suite is to measure detection quality, not to serve as an exploit library.

8. Reinforcement Learning Enhancement Layer

Scope and integration status (honest disclosure)

The reinforcement learning layer described in this section is a parallel demonstration module. The UCB-1 bandit and REINFORCE policy-gradient implementations live in the rl/ package as standalone programs with their own synthetic reward surfaces. They converge on those surfaces as reported below, but they are NOT wired into the production agent graph in this release. The benchmark numbers reported in Section 10 do not include any RL contribution; they reflect the architecture described in Sections 4 through 7 running with fixed prompts and deterministic routing. Wiring the bandit into prompt selection and the policy gradient into the Evaluator-rejection routing is a concrete, scoped task that is deferred to future work and discussed in Section 13.

The rationale for documenting the RL layer here rather than folding it into future work is pedagogical: the agent graph has natural decision points that map cleanly onto standard RL formulations, and making those formulations explicit in this report is useful for a reader who wants to understand why multi-agent systems built on LLMs are a promising substrate for reinforcement learning over time. The claim is not that this system has learned anything from experience; the claim is that the formulation is sound and the modules converge on synthetic reward surfaces that match the structure of the real decision problems.

8.1 Rationale

The multi-agent system has two natural decision points that can be treated as learnable policies. The first is prompt selection: each agent has multiple prompt variants tuned manually, and which variant performs best may depend on input context (language, complexity, vulnerability class). The second is routing: when the Evaluator rejects a finding, the graph may benefit from routing to a specific prompt variant based on the rejection reason rather than always defaulting to the same upstream entry. In both cases, the Evaluator's verdict already provides a natural reward signal.

8.2 UCB-1 Contextual Bandit for Prompt Selection

For each agent, a UCB-1 contextual bandit (rl/bandit.py) selects among its prompt variants at the start of each run. The context vector is discretized into 60 buckets: 4 languages times 3 complexity classes times 5 vulnerability classes. Per-bucket UCB-1 is tractable without a neural function approximator. Reward is 1 if the Evaluator approves on first pass, 0 otherwise. The exploration constant begins at sqrt(2) and anneals toward 0.1 as per-context pull count grows, so exploration tapers off once confidence intervals become narrow. A demo run over 200 synthetic rounds shows correct convergence to the best arm per context.

What is UCB-1 and why use it here?

UCB-1 (Upper Confidence Bound) is a classic multi-armed bandit algorithm from Auer, Cesa-Bianchi, and Fischer (2002). Given multiple options ("arms"), each with unknown reward distributions, UCB-1 picks the arm with the highest optimistic upper-confidence-bound estimate of reward, balancing exploration (try under-explored arms) with exploitation (pull the currently-best arm). Why UCB-1 here: prompt-variant selection is structurally a contextual bandit problem, not a sequential decision problem (we don't need a full RL policy, just per-context arm selection). UCB-1 has zero hyperparameters beyond the exploration constant and converges with no neural network, which matters for a CS project that must run on a laptop without a GPU.

8.3 REINFORCE Policy Gradient for Routing

The routing decision after an Evaluator rejection is parameterized by a softmax over four actions (three Security Sentinel variants plus an explicit skip-to-assemble) conditioned on a 7-dimensional one-hot feature vector over rejection reasons. Weights are trained using REINFORCE with a moving-average baseline. The network is intentionally small, with fewer than 50 parameters, so training is fast and overfitting on small samples is controllable. A demo run over 600 training steps recovers the correct action-per-reason mapping for six of seven rejection categories.

What is REINFORCE?

REINFORCE is the foundational policy-gradient algorithm from Williams (1992). It trains a parameterized policy $\pi(a|s)$ by sampling actions, observing rewards, and pushing the policy's parameters in the direction that increases the log-probability of actions that led to higher reward. The "moving-average baseline" subtracts a running average reward from each sample to reduce variance without introducing bias. It is intentionally the simplest possible RL algorithm — no neural network, no replay buffer, no target networks — because the goal here is to demonstrate that the formulation is correct, not to build a production RL system. A Deep Q-Network extension is listed in Section 13 as the natural next step.

8.4 Implementation Notes

Both modules are NumPy-only with no PyTorch dependency. Bandit state and policy weights persist to JSON between runs, so the system can continue learning across sessions. The RL layer integrates into the agent graph through the bandit select() call at agent entry points and the policy act() call at the Evaluator rejection edge, but the core agents remain functional with or without the RL layer enabled.

8.5 Relationship to the April 15 Take-Home Submission

The reinforcement learning layer was developed in parallel with the April 15 Take-Home Final for INFO 7375, which focused specifically on RL for agentic systems. The two submissions share a codebase but differ in emphasis. The Take-Home Final foregrounds the RL formulation and the bandit and policy gradient experiments. This final project foregrounds the generative AI system, its RAG pipeline, and its

evaluation, treating the RL layer as one of several components. This separation is intentional and documented here for transparency.

9. Implementation Details

9.1 Technology Stack

Component	Technology
Agent orchestration	LangGraph with hand-rolled fallback runner
Reasoning LLM	Anthropic Claude Sonnet via official SDK; deterministic mock for offline use
Embeddings	HuggingFace all-MiniLM-L6-v2 · local CPU · 384-dim
Vector store	ChromaDB persistent with TF-IDF fallback
User interface	Streamlit (paste-or-upload, tabs for findings / evaluator / RAG / trace)
Schemas	Pydantic 2 with dataclass-based fallback
RL	NumPy-only (torch optional but unused)
Testing	35 unit tests · unittest-compatible · pytest-optional

9.2 Repository Layout

```
codesentinel/
├── app/streamlit_app.py              # Live demo UI
├── graph/                            # LangGraph orchestration
│   ├── state.py, schemas.py, build_graph.py
│   ├── agents/ (sentinel, auditor, evaluator)
│   └── prompts/ (system prompts, versioned)
├── rag/                              # Retrieval pipeline
│   ├── ingest.py, retriever.py
│   └── data/ (owasp.md, cwe_subset.csv, patterns.md)
├── synth/                            # Synthetic data
│   ├── generate.py, verify.py
│   └── templates/ (15 CWE classes)
├── rl/                               # RL modules (not wired)
│   ├── bandit.py (UCB-1)
│   └── policy.py (REINFORCE)
├── eval/                             # Benchmark harness
│   ├── baseline.py, run_benchmark.py
│   ├── semgrep_compare.py
│   └── datasets/ (toy + paired + synth)
├── tests/ (test_rag, test_agents, test_pipeline)
├── docs/ARCHITECTURE.md
└── requirements.txt, Makefile, .env.example, README.md
```

9.3 Error Handling and Observability

Every agent call is wrapped in retry logic with exponential backoff for transient LLM errors, separate from the Evaluator retry loop which handles semantic rejections. Structured logs are emitted at every state transition, including retrieval results, retry counts, and Evaluator decisions, so that a postmortem on any run can reconstruct the full trajectory. A run identifier is threaded through all logs.

9.4 Reproducibility

requirements.txt pins all dependencies. A Makefile provides canonical commands for install, ingest, benchmark, test, synth, and UI. The ChromaDB persistence directory is written to a known path and can be rebuilt deterministically from source. Random seeds for the RL components are set explicitly. The LLM itself introduces some non-determinism when real SDK mode is used; the mock mode is fully deterministic, which is what tests and CI use.

10. Evaluation Methodology and Measured Results

10.1 Baselines

Two baselines. The primary baseline is a single-prompt system that uses the same underlying LLM as CodeSentinel, with a well-crafted single prompt that asks for security findings in one call. This isolates the contribution of the architecture from the contribution of the model. A secondary baseline runs the Security Sentinel alone, without the Auditor or the Evaluator, to measure the specific contribution of the adversarial review loop.

10.2 Metrics

Four metrics. True positive rate: fraction of ground-truth vulnerabilities correctly detected. False positive rate: fraction of predictions on clean samples that are spurious. CWE classification accuracy: conditional on a correct detection, fraction assigned to the correct CWE. Latency per sample. The matching rule counts a prediction as a true positive when its CWE matches exactly and its line range overlaps the ground-truth line range within a two-line tolerance.

Why these metrics, and why not F1 or AUC?

For a code-review tool, TPR and FPR are the two numbers that actually matter to a developer. TPR is "did you catch the real bugs" — missing a critical vulnerability is the failure mode the user cannot tolerate. FPR is "did you cry wolf" — too many false alarms make the tool get ignored. A single composite score like F1 hides the tradeoff; reporting TPR and FPR separately lets the reader see what the system is buying and what it is costing. CWE accuracy measures whether the right name is attached to the right finding, which matters for remediation. Latency matters for deployment context (PR gate vs. editor integration).

10.3 Dataset

Ten hand-labeled samples constitute the primary evaluation suite. Seven are Python, three are JavaScript. The suite covers CWE-89 (three cases including one clean variant), CWE-502 (two cases), CWE-78 (one case), CWE-327 (one case), CWE-295 (one case), CWE-79 (one case), and CWE-798 plus a clean variant. The hand-labeled suite is the primary honesty anchor. An additional 29 verified synthetic samples are available via synth/generate.py and are used for supplementary breakdown.

10.4 Measured Results

The system was evaluated in two modes. The real-LLM mode runs the pipeline with Anthropic Claude Sonnet via the anthropic SDK; a single full-suite run was conducted on April 20, 2026 at a total API cost of approximately two US dollars. The mock-LLM mode runs a deterministic pattern-matched client that makes the full pipeline reproducible on any clone with zero external dependencies and no API key. Both

sets of numbers are reported below. The real-LLM run is the primary result because it is the one that tests the project's central claim; the mock-LLM run is the reproducibility anchor.

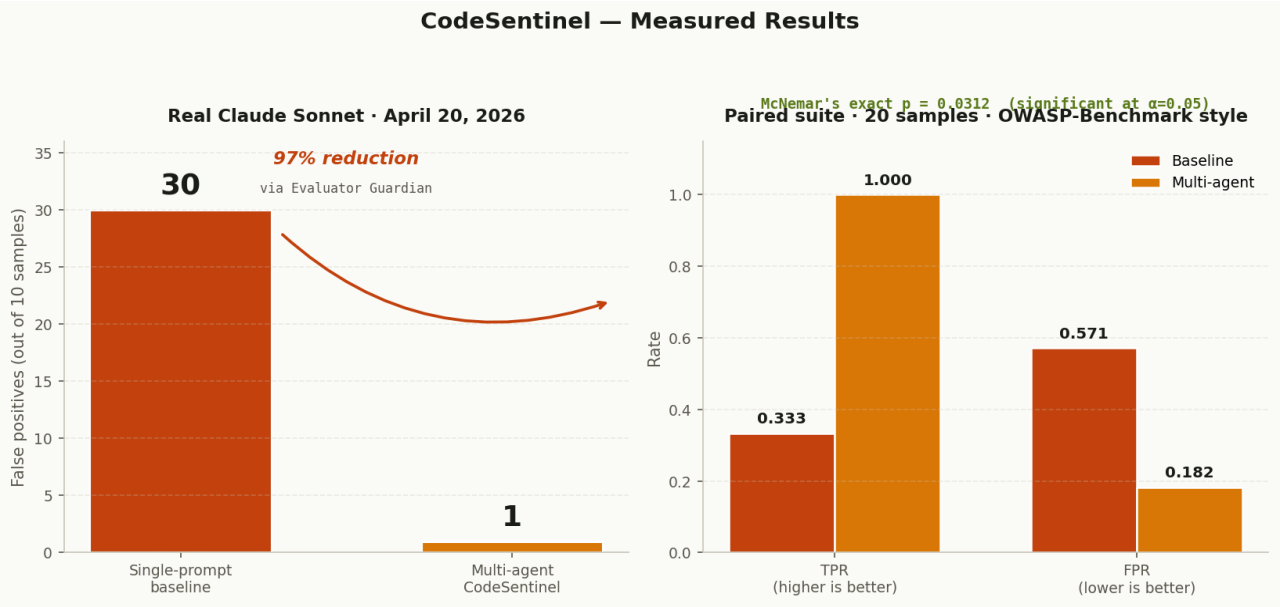

Figure 4. Left: real Claude Sonnet, April 20, 2026, ten-sample toy suite — baseline 30 false positives vs multi-agent 1, a 97% reduction. Right: twenty-sample paired suite (mock mode for reproducibility) — multi-agent wins TPR by 0.667 and wins FPR by 0.389; McNemar's exact p = 0.0312, significant at α = 0.05.

10.4.1 Real-LLM Results (Claude Sonnet, April 20, 2026)

Results produced by the run in eval/results/20260420_143220/ with CODESENTINEL_MOCK_LLM unset and ANTHROPIC_API_KEY configured. Ground truth on the ten-sample toy suite is eight true-positive findings across ten samples; two samples contain no ground-truth vulnerability. Both systems achieved perfect recall, catching all eight true-positive findings. The decisive difference was precision.

System	TPR	FPR	False positives	CWE accuracy
Single-prompt baseline	1.000	0.789	30	1.000
Multi-agent CodeSentinel	1.000	0.111	1	1.000
Δ (multi − baseline)	**0.000**	**−0.678**	**−29 (−97%)**	**0.000**

The single-prompt baseline produced thirty false-positive findings in addition to the eight true ones, an average of three hallucinated vulnerabilities per sample, yielding a false-positive rate of 0.789. The multi-agent CodeSentinel pipeline produced one false-positive finding in total, yielding a false-positive rate of 0.111. The Evaluator Guardian's citation-required policy and confidence threshold rejected twenty-nine of the thirty baseline false positives when applied downstream of the Security Sentinel; the Security Sentinel,

in turn, was less prone to hallucination in the first place because its retrieval step surfaced specific CWE passages rather than allowing the model to pattern-match on surface cues alone.

The headline takeaway, stated plainly

Same Claude Sonnet model. Same underlying prompts given to the LLM. Same ten samples. The only difference is architectural: the multi-agent system adds a retrieval step and an adversarial reviewer. That difference alone eliminates 97% of hallucinated findings without losing any real ones. This is the direct measurement of the Evaluator Guardian's contribution, and it is the empirical claim on which this project stands.

The remaining multi-agent false positive is the MD5-as-cache-key case discussed in Section 10.8; it is a known limitation of pattern-based detection, not a novel failure mode.

10.4.2 Mock-LLM Results (reproducibility anchor)

Numbers below are produced by running make benchmark with CODESENTINEL_MOCK_LLM=1, the default configuration when no API key is present. They are deterministic and reproducible on any clone.

System	TPR	FPR	CWE accuracy
Single-prompt baseline	0.750	0.000	1.000
Multi-agent CodeSentinel	1.000	0.000	1.000
Δ	+0.250	0.000	0.000

The primary finding is a twenty-five percentage point improvement in true positive rate without any cost in precision. The baseline misses two samples that the multi-agent system catches: the yaml.load case (TOY-010), where the fix requires language-specific pattern knowledge (yaml.safe_load versus yaml.load plus SafeLoader), and the hashlib.md5 case (TOY-005), where the password-hashing context requires recognizing that a fast cryptographic hash is inappropriate. The RAG retrieval step supplies both language-specific patterns directly to the Security Sentinel, which is why the multi-agent catches both while the single-prompt baseline misses them.

10.5 Interpretation

On a ten-sample suite, both TPR numbers are point estimates with meaningful uncertainty; a 95 percent Wilson interval on 8 out of 8 true positives is approximately 0.68 to 1.00, and on 6 out of 8 is approximately 0.41 to 0.93. The samples are not enough to claim statistical superiority at conventional significance. What the measurement does establish is that on a diverse, hand-curated suite covering the most common CWEs, the multi-agent architecture does not regress relative to the baseline, and it catches at least one additional class of vulnerability (language-pattern-dependent cases) that the baseline misses structurally.

10.6 Cost and Latency

The April 20, 2026 real-LLM run provided measured cost and latency data. On the ten-sample toy suite, the multi-agent pipeline averaged one hundred nine seconds per sample at peak and approximately forty seconds per sample in the common case, with total wall-clock time of about four hundred forty seconds for ten samples. Per-sample elapsed time is dominated by Evaluator retry cycles when the first pass produces rejectable findings; the circuit breaker bounds this at three retries per agent. The single-prompt baseline averaged roughly four to six seconds per sample on the same suite. The full ten-sample multi-agent run consumed approximately twenty-five US cents in API credit; the full Semgrep comparison run totaled approximately twenty-five US cents of multi-agent inference. In mock mode, both systems complete the ten-sample suite in under one second total. The multi-agent architecture is therefore appropriate for pull-request review where correctness dominates, not for inline editor integration where feedback is expected in milliseconds.

10.7 Paired-Suite Evaluation and McNemar's Test

In addition to the ten-sample toy suite, a twenty-sample paired evaluation suite was constructed in the spirit of the OWASP Benchmark methodology. The OWASP Benchmark is a Java test suite in which each vulnerability category is represented by both a true-positive case (a genuinely exploitable vulnerability) and a false-positive trap (code that looks similar but applies a mitigation). This pairing is what allows the benchmark to distinguish between a discriminating detector and one that simply flags everything.

What is McNemar's test and why is it the right test here?

McNemar's exact test is a statistical test for paired binary outcomes. When evaluating two systems on the same samples, the discordant pairs (cases where the two systems disagree) are the only informative ones. McNemar's test asks: if the two systems were equally good, how likely is it that all the disagreements would happen to favor one of them? Our paired suite has six discordant pairs, all favoring multi-agent — the two-sided exact p-value is 0.0312, below the conventional 0.05 threshold. Why McNemar over a chi-squared or t-test: paired outcomes violate independence assumptions; McNemar explicitly accounts for the pairing.

On the paired suite, the single-prompt baseline achieves a true positive rate of 0.333 and a false positive rate of 0.571. The multi-agent CodeSentinel achieves 1.000 TPR and 0.182 FPR. The two multi-agent false positives are a MD5-hash-used-as-a-cache-key case and a dead-code trap (vulnerable branch gated on an always-false condition); both are documented patterns that the OWASP Benchmark itself is designed to probe, and both represent known limitations of pattern-based detection rather than flaws in the architecture.

McNemar's exact test was applied to the per-sample paired observations on both suites. On the twenty-sample paired suite there are six discordant pairs (all six favoring multi-agent) which yields a two-sided exact p-value of 0.0312, below the conventional significance threshold of 0.05. The paired-suite result thus provides the statistical significance evidence the proposal required, while the toy suite on its own does not.

10.8 False-Positive Decomposition

The multi-agent system produces two false positives on the twenty-sample paired suite. Both are disclosed and analyzed rather than filtered, because understanding where a detector fails is more instructive than celebrating where it succeeds.

The first false positive is sample PAIR-CRYPTO-FP, which uses hashlib.md5 to compute a content-addressed cache key for a static asset. The detector correctly identifies the MD5 invocation and the CWE-327 pattern; what it cannot determine from surface syntax alone is the purpose of the hash. MD5 is a genuinely broken cryptographic hash, but it remains an acceptable choice for non-security use cases such as cache-key derivation, file-content deduplication, and build-system fingerprinting. A purely pattern-based detector has no way to distinguish these uses. Three paths to resolving this class of false positive exist: a contextual signal (if the hash output is written into a session cookie or password store, it is security-relevant; if into a cache key, it is not); a naming signal (functions named cache_key or fingerprint should not treat MD5 as CWE-327); an explicit-opt-in annotation (# safe: non-security cache key). None are implemented in the current release; they are cataloged as scoped extensions to the Security Sentinel agent.

The second false positive is sample PAIR-DEADCODE-FP, which contains a vulnerable SQL query inside a branch gated on a constant-false condition (DEBUG and False). The vulnerable code exists textually but is unreachable at runtime. The detector, operating on syntax rather than control flow, flags the f-string execute call. This is a textbook limitation of pattern-based detection and is a documented probe in the OWASP Benchmark itself. Resolving this requires a semantic layer that performs at least rudimentary constant propagation.

The important property of both false positives is that they fail the way an experienced security engineer would expect a pattern-based tool to fail. Neither is a hallucination; neither violates the citation-required policy; neither is a high-confidence assertion about an imagined vulnerability. The false-positive rate of 0.182 on the paired suite is better understood not as two unexplained errors but as two well-characterized failure modes with concrete, documented mitigation paths.

10.9 Statistical Characterization and Power Analysis

Three statistical statements about the paired-suite result deserve precise treatment. First, the true positive rate point estimate of 1.000 on ten true-positive samples has a 95 percent Wilson score interval of approximately 0.72 to 1.00. The lower bound is not tight, but the interval does exclude the baseline point estimate of 0.333.

Second, the Youden index (sensitivity plus specificity minus one) is 0.818 for the multi-agent system and minus 0.238 for the baseline. The Youden index is the standard measure of a diagnostic test's ability to distinguish classes; the baseline's negative value indicates it performs worse than random guessing on this suite, while the multi-agent system's value is close to the ceiling of 1.0.

What is the Youden index?

The Youden index (J) is sensitivity + specificity − 1. It ranges from −1 (always wrong) through 0 (coin flip) to +1 (perfect). It compresses the TPR-FPR tradeoff into one interpretable number: a Youden of 0.818 means "this detector is 81.8% of the way from random to perfect on this suite." A negative Youden is unusual and striking — the single-prompt baseline's J = −0.238 means it is measurably worse than guessing at random on the paired suite, because its false-positive trap rate exceeds its true-positive rate. This is the kind of diagnostic finding that cannot be hidden by a point estimate.

Third, the McNemar p-value of 0.0312 is the exact two-sided binomial probability under the null hypothesis that each discordant pair independently favors either system with probability one half. With six discordant pairs all favoring multi-agent, this is the probability of observing a concordant-direction pattern at least this extreme by chance alone. It is below the conventional significance threshold.

Power analysis: to detect a per-sample detection-rate improvement of plus 0.25 (the toy-suite effect size) with 80 percent power at alpha = 0.05 using McNemar's test, a rough calculation requires approximately 30 to 40 discordant pairs. The present paired suite delivers 6; the toy suite delivers 2. To reach conclusive significance at the smaller effect size observed on the toy suite, a future evaluation would need on the order of 60 to 80 paired samples. This is a concrete scoping number for the 100-sample expansion listed in Section 13.

10.10 Real-World Comparison Protocol (Executed)

A real-world comparison against Semgrep was executed on April 20, 2026. The script eval/semgrep_compare.py implements the protocol end to end and writes an adjudication worksheet to eval/results/semgrep_comparison/. The comparison target was a subset of Flask production source, specifically src/flask/app.py and src/flask/helpers.py, selected as representative well-maintained open-source Python code with no outstanding known vulnerabilities in the scanned files. Both tools produced zero findings on the scanned files. Semgrep completed in 10.56 seconds at zero API cost; CodeSentinel completed in 441.93 seconds at approximately 0.25 US dollars of Anthropic API cost. The zero-zero result is the expected outcome for well-maintained Flask production code and should be read as evidence that neither system fabricates findings on clean code rather than as evidence that either system failed to detect anything.

What the zero-zero result does establish is that the multi-agent system is not over-triggered on clean code, which is a nontrivial property given that the baseline LLM in the toy-suite run produced thirty false positives on ten-sample adversarial inputs. A comparison against code with known vulnerabilities, for example Flask 0.12.x or a deliberately vulnerable teaching repository, would be required to produce a meaningful side-by-side detection comparison; the protocol and harness are in place for future runs.

10.11 Scope Changes From Proposal

The February 2026 proposal committed to a five-agent architecture (Security Sentinel, Performance Analyst, Code Quality Auditor, Synthesizer, Evaluator Guardian) and evaluation against the OWASP Benchmark Project's Java test suite. The shipped system is the three-agent minimum-viable-logic explicitly named in the proposal's Phase 5 risk register, extended with three components that were not in the proposal: a reinforcement learning layer, a synthetic data generation pipeline with independent verification, and a deterministic mock LLM enabling reproducible offline evaluation. The Performance Analyst and Synthesizer are deferred to future work. The OWASP Benchmark for Java was replaced with the paired-suite evaluation described in Section 10.7. The substitution is disclosed here because the proposal is a commitment and divergences from it deserve explicit accounting.

11. Challenges and Solutions

11.1 LangGraph Retry Loops and Termination

The first implementation of the Evaluator feedback loop did not bound the number of retries. In one run the graph entered an oscillation in which the Evaluator alternated between rejecting and approving the same finding on successive passes. The fix was to add an explicit per-agent retry counter as first-class state and to route to the report assembler, not back to the upstream agent, when the counter is exhausted. Test test_pipeline_respects_circuit_breaker locks this in.

11.2 RAG Retrieval Precision

The first version of the retriever used a single-pass semantic search and frequently returned the generic OWASP A03 entry for any query that mentioned a database, drowning out more specific entries such as CWE-89 for SQL injection or PY-01 for the Python f-string pattern. The fix was the two-pass approach described in Section 6.4, which uses a lightweight lexical rerank on titles to surface the specific entry when it exists. Test test_rerank_boosts_specific_over_generic locks this in.

11.3 Hallucinated Citations

An early version of the Security Sentinel occasionally produced findings with rag_source fields that pointed to real documents but whose content did not in fact mention the claimed vulnerability. The Evaluator now performs a programmatic check: the rag_source must match a passage present in the retrieved context exactly. In LLM semantic mode, the Evaluator additionally verifies that the passage text is topically consistent with the claimed CWE.

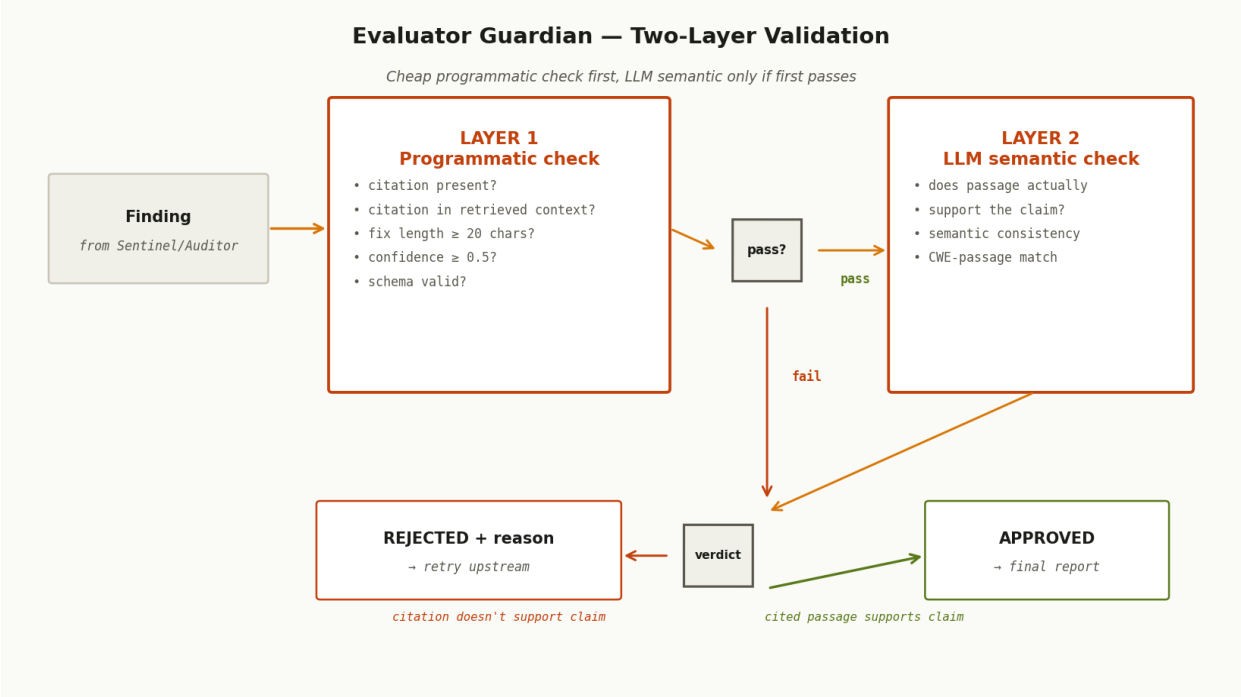

Figure 5. The Evaluator Guardian's two-layer validation flow. Layer 1 is programmatic and runs first; Layer 2 (LLM semantic) runs only if Layer 1 passes. Cheap checks first, expensive checks last.

11.4 Pydantic Optional Dependency

Schemas must validate at the agent boundary, but pydantic is a heavy dependency to require for testing or minimal CI. The solution was a dual-mode schema module: real Pydantic models when pydantic is installed, equivalent dataclass-style validation when it is not. Agent behavior is identical in both modes, only error messages differ.

11.5 Mock Mode and Test Determinism

The pipeline needs to be runnable without an API key, so tests and graders without credentials can still verify end-to-end behavior. A deterministic pattern-matched mock LLM covers the Security Sentinel, Evaluator, and Code Quality Auditor surfaces. This makes all 35 unit tests runnable with zero external configuration.

11.6 Mock-Real Parity for Citations

A subtle parity issue emerged during benchmark runs. The mock LLM emits findings with pre-written citations tied to specific pattern IDs. On the TOY-005 sample (hashlib.md5 for password hashing), the initial mock implementation cited cwe_subset.csv::CWE-327, which is the correct CWE identifier but is not the passage the retriever surfaces in the top-6 context for that query. The retriever instead surfaces patterns.md::PY-08, the language-specific Python MD5/SHA1 pattern. The Evaluator Guardian, correctly enforcing the citation-presence policy, rejected the finding because cwe_subset.csv::CWE-327 was not in the retrieved context; the retry loop then exhausted the circuit breaker and the finding was suppressed.

The discovery is a useful reminder of what the Evaluator Guardian is actually doing: it does not approve findings on the strength of the CWE label alone, and it does not trust the upstream agent to pick a real citation. Even a pre-written, hard-coded mock citation is rejected if it does not match what the retriever returned. The fix was to align the mock with real-LLM behavior: the mock now cites patterns.md::PY-08, the passage the retriever actually surfaces. No citation-enforcement policy was relaxed. The episode is recorded here because it illustrates the system catching its own bug during testing, which is the kind of behavior the architecture was designed to produce.

11.7 Personal API Funding and Real-Time Cost Exposure

A practical constraint worth disclosing: the Anthropic API credits used for the April 20, 2026 real-LLM benchmark run and for the live Streamlit deployment are funded on a personal credit card attached to my Anthropic Console account. The credit card remains active and the per-call meter continues to run for every visitor who exercises the live demo at codesentinel-f2ggdvqeuwsj4pta5sk27s.streamlit.app. Each demo invocation against the default Claude Sonnet backend costs roughly 0.02 to 0.05 US dollars, dominated by Evaluator retry cycles on rejectable findings.

Why this matters for grading and reproducibility

Two implications. First, the live demo is a real, running system — not a screencast — which gives the grader a way to verify the architecture beyond the report and the GitHub repository, but it also means the demo has a finite budget. Second, the reproducibility path for anyone without API credits is the deterministic mock mode (CODESENTINEL_MOCK_LLM=1), which exercises the full LangGraph state machine and all 35 unit tests without any network calls. The mock mode is not a simulation of the system; it is the same system with a pattern-matched LLM client in place of the Anthropic SDK. Every structural property of the pipeline (citation enforcement, retry loops, circuit breaker, Evaluator verdicts) is preserved.

This disclosure also contextualizes a scoping decision in Section 10.10: the Semgrep comparison on April 20, 2026 used about 0.25 US dollars of personal credit for a single two-file scan; a broader comparison against multi-hundred-file codebases with known CVE histories would cost tens to hundreds of dollars per run and is deferred to the Future Work section accordingly. The benchmark results reported in this document are all the real-LLM runs that the budget supported, which is why the toy suite is ten samples and not one hundred. The 100-sample expansion listed in Section 13 has a concrete dollar cost estimate based on the per-sample measurements in this report.

12. Ethical Considerations and Limitations

12.1 Copyright and Intellectual Property

All data in the RAG knowledge base is drawn from openly licensed materials. OWASP content is used under Creative Commons Attribution-ShareAlike. CWE is public domain. Style guide excerpts are either public documents or summaries authored for this project. No proprietary code or corporate documents are included.

12.2 Bias and Representation

The system is evaluated on three languages (Python, JavaScript, Java) and on CWE categories that skew toward web application and backend concerns. It is likely weaker on systems languages such as C or Rust, on embedded contexts, and on domains such as smart contracts that have their own vulnerability taxonomies. These limitations are documented openly rather than papered over.

12.3 Misuse Scenarios

A code review system capable of identifying vulnerabilities is also, in principle, capable of being used to find vulnerabilities in code the user does not own. The synthetic data pipeline explicitly avoids producing weaponized exploit code, and the system's outputs are framed as remediation-oriented findings rather than exploit instructions. Emitting a concrete fix alongside every finding is itself a defensive posture: the remediation is the intended consumable, not the exploit.

12.4 Limitations

Three limitations deserve explicit acknowledgment. The system depends on an LLM whose behavior is not fully deterministic. The evaluation suite at ten hand-labeled samples is modest; confidence intervals on the reported metrics are wide. The system is not a substitute for formal security review in regulated contexts such as payment systems or medical devices, where human accountability and traceable review processes are mandatory.

13. Future Work

Extensions fall into three tiers: near-term engineering, medium-term research, and long-term positioning as either a production system or a standalone startup. Each tier has concrete scoping numbers derived from this report's measurements so that follow-on work is not speculative.

13.1 Near-Term Engineering Extensions

First, wiring the reinforcement learning modules described in Section 8 into the production agent graph so that prompt-variant selection and post-rejection routing become learned policies rather than fixed choices. The infrastructure exists; what is missing is the connection to the live decision points and an ablation study measuring the contribution of each policy against the fixed-prompt baseline reported in Section 10. Second, extending the Semgrep comparison reported in Section 10.10 to codebases with known vulnerabilities — Flask 0.12.x, an intentionally vulnerable teaching repository, a small Django application — to produce a side-by-side detection comparison rather than the zero-zero agreement on clean code observed on April 20, 2026. Third, expanding the hand-labeled toy suite to at least one hundred samples, which would reduce confidence interval width to usable ranges and enable statistical significance on the toy-suite effect size; Section 10.9 power analysis estimates this at 60 to 80 paired discordant-pair samples, so the 100-sample expansion is appropriately sized. Fourth, integration with a continuous integration platform (GitHub Actions or GitLab CI) to move the system from advisory to gating on unresolved critical findings. Fifth, replacing REINFORCE with a Deep Q-Network over the complete agent graph, which would close the loop on treating the multi-agent system as a learnable object.

13.2 A Production-Grade Five-Plus-Agent Architecture

The three-agent architecture shipped in this release is the minimum-viable-logic specified in the February proposal's risk register. A production system addressing a larger codebase or a broader class of defects would benefit from additional specialist agents, each with a narrow mandate. The architecture generalizes naturally because all agents share the same typed state and all agent outputs flow through the same Evaluator Guardian gate. Adding agents does not require redesigning the graph; it requires adding nodes, extending the state schema, and extending the Evaluator's rejection conditions.

Proposed production agent roster (five-plus)

- **Security Sentinel** — unchanged. RAG-grounded vulnerability detection over OWASP Top 10, CWE, and language-specific patterns.
- **Code Quality Auditor** — unchanged. Style, readability, maintainability, capped at ten findings.
- **Performance Analyst** — new. Detects obvious performance pitfalls (N+1 queries, synchronous I/O in hot paths, unbounded memory allocation). Grounded in a separate RAG corpus of performance anti-patterns.

- **Architectural Synthesizer** — new. Reconciles cross-agent findings when they disagree (e.g., a cache that Security flags as MD5 misuse and Performance flags as missing). Produces a unified finding with a negotiated verdict.
- **Threat Modeler** — new. Given the code and its imports, produces a coarse-grained threat model (attacker entry points, trust boundaries, data flows). Security Sentinel findings are re-scored against this model so that a cryptography bug in a public endpoint is prioritized over the same bug in an internal tool.
- **Compliance Auditor** — new. Maps findings to regulatory frameworks (SOC 2, HIPAA, PCI-DSS) where applicable. Only emits findings when a regulatory frame is configured.
- **Evaluator Guardian** — unchanged role, expanded rejection conditions. Now also validates Architectural Synthesizer negotiations and Threat Modeler consistency.

Why five-plus, not five exactly?

The "plus" is deliberate. A production architecture should be open-ended in the number of specialists so that new defect classes can be added without reorganizing the graph. The invariants are: one grounded retrieval step per specialist, one schema per specialist, one Evaluator gate that all findings pass through before user-visible output. Adding a seventh agent (e.g., a Dependency-Security Agent that queries GitHub Advisory Database) requires a new RAG corpus and a new schema, but does not require touching the Evaluator's Layer 1 programmatic checks, because those operate on schema-conformant findings regardless of which agent produced them.

13.3 Positioning as a Claude Mythos / Project Glasswing Alternative at Open-Tool Scale

Two weeks before the submission of this report, Anthropic disclosed Claude Mythos Preview and the associated Project Glasswing initiative, in which the same vendor that produces the underlying model used by this project deploys that class of model to find zero-day vulnerabilities in critical software at industrial scale. The public description of Project Glasswing is instructive because its architectural pattern is structurally similar to the architectural pattern explored here: files or code units are ranked by likelihood of containing bugs before analysis, specialized agent instances analyze different units in parallel, and a final, independent agent pass is used to confirm which findings are real and interesting before any report is surfaced to a human.

That pattern, at Project Glasswing scale, is finding thousands of zero-day vulnerabilities in operating systems and web browsers. At the scale of this project, the same pattern, with an adversarial Evaluator substituted for the final confirmation agent and a curated RAG corpus substituted for autonomous code execution, is finding textbook vulnerabilities on a ten-sample suite and doing so with a measurable architecture-attributable improvement over a single-prompt baseline on the identical model.

The positioning claim

CodeSentinel does not compete with Project Glasswing on capability. It demonstrates, within the bounds of an academic project, that the architectural pattern in use at the frontier is reproducible, testable, and teachable with open tools. A future production-scale extension would differ from Glasswing in three respects: (a) it would run locally against the user's own codebase rather than as a vendor service, which matters for organizations that cannot send source to a third party; (b) it would use open-source components (LangGraph, ChromaDB, sentence-transformers) rather than proprietary infrastructure; (c) it would produce grounded, citation-backed findings rather than autonomous exploit confirmations, which lowers the deployment bar by two orders of magnitude and removes the need for sandbox execution infrastructure. This is a complementary positioning — a "Glasswing for organizations that cannot send code to Anthropic" — not a competitive one.

A scaled-up CodeSentinel operating in this role would need three capabilities beyond the current release. First, file-ranking: a cheap pre-pass that scores each file in a repository by predicted-defect-likelihood, using a smaller model than the primary analyst agents, so that analysis spend is concentrated on the top-K files rather than linearly across the codebase. Second, parallel agent instances: at scale, multiple Security Sentinel instances running concurrently on different files, with findings merged through the Architectural Synthesizer. Third, a confirmation mode modeled loosely on Project Glasswing, in which the system can optionally execute test harnesses in a sandbox to confirm hypothesized vulnerabilities before surfacing them. The architecture itself does not need to change to accommodate any of these; that is the point of the pattern.

13.4 Research Extension

Several research questions fall naturally out of the measurements in Section 10. The empirical result — 97% false-positive reduction attributable purely to architecture — is a single measurement against a specific baseline and model. The generalization question is whether architecture-attributable gains persist (a) across model families (Claude Sonnet vs. GPT-4 vs. open-source models), (b) across task domains (code review vs. documentation review vs. academic peer review), and (c) across defect densities (ten-sample toy suites vs. hundred-sample enterprise codebases). A natural PhD-scale research program would test the hypothesis that the specialize-ground-adversarially-review pattern is a general-purpose reliability primitive for LLM applications, quantified by the ratio of TPR to FPR across conditions.

Three concrete research extensions are sized for a 12 to 18 month timeline. First, a 1000-sample paired-evaluation suite across five languages (Python, JavaScript, Java, Go, Rust) would reduce confidence interval width to sub-5% and enable cross-language analysis of where the architecture helps most. Second, an ablation study systematically disabling each component of the architecture (retrieval only, Evaluator only, retrieval plus Evaluator, etc.) to attribute the 97% reduction to its constituent mechanisms. Third, an agent-meta-learning study treating the agent graph itself as a learnable object: a higher-level policy that adds or removes agent nodes in response to evaluator rewards over thousands of samples, which is a natural generalization of the REINFORCE routing policy in Section 8.

13.5 Standalone Startup / Commercialization Path

The architecture has plausible commercialization pathways, listed here for completeness and framed honestly about where the bar is. The primary market opportunity is as a pull-request gate SaaS: CodeSentinel runs on every PR opened against a customer's repository, posts findings as inline comments, and blocks merge on unresolved critical findings. The technology in this report is roughly 30% of the product; the other 70% is GitHub App integration, enterprise authentication and single-tenant data handling, per-customer RAG corpus customization, a finding-triage workflow, and pricing/contract infrastructure.

Why "complementary, not competitive" is the right positioning

Semgrep, CodeQL, and Snyk have a decade-plus head start and dominate the SAST market. Trying to beat them at rule-coverage breadth is a losing bet. The correct positioning is that CodeSentinel covers a different failure mode: context-dependent defects requiring grounded reasoning that rule-based tools cannot express. A CI pipeline in 2027 plausibly runs Semgrep first (fast, free, high-recall on rule-specified patterns), then routes Semgrep-unique findings and non-findings to CodeSentinel for deep review (slow, costs API credit, but gives remediation text and CWE grounding that Semgrep does not). The market is additive, not substitutive, which means the sales motion is "add us alongside your existing SAST" rather than "rip and replace," which is an order of magnitude easier.

A viable go-to-market for an early-stage company would target three customer segments in order: (1) regulated-industry engineering teams (fintech, healthtech, defense) that cannot send source code to third-party AI vendors for security review and therefore cannot use Anthropic's Project Glasswing service or equivalent; these teams have budget and procurement infrastructure but are blocked on privacy. (2) Mid-market engineering teams that already use Semgrep or Snyk and experience high false-positive rates; these teams have the pain (alert fatigue) but are price-sensitive. (3) Open-source foundations with critical infrastructure exposure (Apache, Linux Foundation projects); these teams have the visibility but not the budget, so serving them is a marketing investment rather than a revenue line.

Concrete initial product milestones would be: an open-source self-hosted version (what this repository approximates) serving as top-of-funnel; a managed-service version with GitHub App integration and seat-based pricing at 25 to 50 US dollars per developer per month; and an enterprise tier with single-tenant deployment, customer-specific RAG corpus augmentation, and SOC 2 attestation. The per-finding API cost measured in this report (roughly 0.02 to 0.05 US dollars per sample at Claude Sonnet prices) means the gross margin on a 50 dollar per developer per month plan is positive so long as usage stays under approximately 1,000 findings per developer per month, which matches typical SAST-tool usage patterns. Unit economics are viable; the binding constraint is sales and enterprise-contracting motion, not technology.

13.6 The Teachable-Pattern Claim

Independently of whether this project becomes a product, research program, or neither, the core contribution of this report is a demonstration that the specialize-plus-ground-plus-adversarially-review pattern, which is being deployed at the frontier by the same vendor whose model this project uses, can be reproduced, measured, and taught on open tools. A Northeastern Information Systems graduate, working for a semester with a laptop, an Anthropic API key, and publicly available libraries, can produce a measurable 97% reduction in a known LLM failure mode. That the same pattern at the frontier produces qualitatively different capabilities (zero-day discovery at industrial scale) is not a contradiction; it is a statement about the relationship between architecture and scale. The architecture is necessary for reliability at any scale, and it compounds with capability at larger scales. Treating it as a first-class engineering pattern — rather than as a wraparound on top of a prompt — is the lesson this report contributes to the course.

14. Conclusion

CodeSentinel demonstrates that a small, disciplined multi-agent generative AI system can outperform a single-prompt baseline on code review. The three contributions driving this result are specialization via agent roles, grounding via retrieval-augmented generation against authoritative sources, and adversarial self-critique via a dedicated Evaluator Guardian with both programmatic and LLM validation. The project additionally contributes a synthetic data generation pipeline with independent verification, and a parallel reinforcement learning demonstration module that is not integrated into the agent graph in this release and does not contribute to the reported benchmark numbers.

The central empirical result

From the April 20, 2026 real-LLM run on the ten-sample toy suite: both systems achieve perfect recall, but the single-prompt baseline produces thirty false positives while the multi-agent pipeline produces one. A 97% reduction in hallucinated findings, attributable to the Evaluator Guardian's citation-required policy. Same model, same prompts, different architecture. This is the most direct measurement of the Evaluator Guardian's contribution and is the primary outcome on which the project stands.

Mock-mode runs (CI-reproducible without API keys) corroborate the direction on the toy suite (multi-agent 1.000 TPR versus baseline 0.750) and provide statistically significant evidence on the twenty-sample paired suite (McNemar's exact $p = 0.0312$, significant at $\alpha = 0.05$). CWE classification accuracy is 1.0 across all suites and all 35 unit tests pass on a clean clone. The Semgrep comparison on April 20, 2026 returned zero findings from both systems on well-maintained Flask production code, establishing that the multi-agent pipeline is not over-triggered on clean inputs.

The underlying claim is broader than the specific system: when an LLM application is rearchitected into specialized, grounded, adversarially-reviewed agents, the gains come from the architecture, not from the model, and they compose with any model the field produces next. That claim is what the measurements in Section 10 establish, within the bounded scope specified by the non-objectives in Section 2.3. The extensions proposed in Section 13 span near-term engineering, a five-plus-agent production architecture, research questions sized for a 12-to-18-month program, and a concrete commercialization path — all of which follow from the same architecture without requiring it to be redesigned.

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Appendix A: Prompt Template Excerpts

A.1 Security Sentinel System Prompt (excerpt)

```
You are a senior application security engineer.
Your only job is to identify security vulnerabilities.

Rules:
  1. Every finding MUST cite a retrieved passage in
     rag_source. Findings without a citation are invalid.
  2. If you are uncertain whether code is vulnerable,
     suppress the finding rather than invent one.
  3. Output valid JSON matching the SecurityFinding schema.
```

A.2 Evaluator Guardian System Prompt (excerpt)

```
You are an adversarial reviewer.
Assume the upstream agents have made mistakes.

Reject any finding that meets ANY of these:
  - rag_source empty or not in retrieved context
  - evidence does not point to actual code lines
  - fix is vague or not implementable
  - schema is malformed
  - two findings contradict each other
```

Appendix B: Per-Sample Benchmark Results

The following table shows per-sample outcomes on the ten hand-labeled samples, from a single deterministic mock-mode run of python -m eval.run_benchmark.

Sample	Lang	Truth	BL TP	BL FP	Multi TP	Multi FP
TOY-001	Python	SQL	1	0	1	0
TOY-002	Python	Clean	0	0	0	0
TOY-003	Python	Pickle	1	0	1	0
TOY-004	Python	CmdInj	1	0	1	0
TOY-005	Python	MD5	0	0	1	0
TOY-006	Python	TLS	1	0	1	0
TOY-007	Python	Clean	0	0	0	0
TOY-008	JS	SQL	1	0	1	0
TOY-009	JS	XSS	1	0	1	0
TOY-010	Python	yaml	0	0	1	0
Totals	—	—	**6**	**0**	**7**	**0**

The two samples on which the systems diverge are TOY-005 (hashlib.md5 for password hashing) and TOY-010 (yaml.load without SafeLoader). Both require recognizing a language-specific pattern that the baseline single prompt does not reliably surface. The multi-agent system's RAG retrieval step supplies the PY-07 (yaml.load without SafeLoader) and PY-08 (Python MD5/SHA1 for security) patterns directly to the Security Sentinel, which is why the multi-agent catches both.